Accelerated Failure Time Models

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Introduction

Accelerated failure time (AFT) models express covariate effects directly on the *time scale*: each covariate scales the expected survival time by a multiplicative factor. The model is $\log T = x^\top \beta + \sigma W$ with W a standardised noise (extreme-value, Normal, or logistic, depending on the family). The exponentiated coefficient $\exp(-\beta_j)$ is an "acceleration factor" — the multiplier on the time scale per unit increase in x_j. AFT is the natural framework when communicating effects to clinical audiences who reason in survival times rather than hazards, and it remains valid under non-proportional hazards where Cox regression fails.

Prerequisites

A working understanding of parametric survival distributions (Weibull, log-Normal, log-logistic) and the difference between hazard-ratio and acceleration-factor interpretations.

Theory

Common AFT families:

- **Weibull**: log-time errors are extreme-value distributed; this is the unique family with both PH and AFT interpretations, with $\beta_{\mathrm{PH}} = -k\beta_{\mathrm{AFT}}$ where k is the Weibull shape.
- **Log-Normal**: errors are Normal on the log-time scale; produces non-monotone hazards.
- **Log-logistic**: errors are logistic on the log-time scale; produces non-monotone hazards with heavier tails than log-Normal and a closed-form quantile function.
- **Generalised Gamma**: a three-parameter family that nests Weibull, log-Normal, and exponential; useful for selecting among nested AFT alternatives via likelihood-ratio tests.

The acceleration-factor interpretation is direct: $\exp(\beta_j) = 0.7$ means survival times are 30 % shorter for each unit increase in x_j; $\exp(\beta_j) = 1.3$ means 30 % longer.

Assumptions

The chosen parametric family is correct (verifiable through residual diagnostics and AIC comparisons), independent non-informative censoring, and the absence of unmodelled time-varying covariate effects.

R Implementation

```r
library(survival); library(flexsurv)

data(lung)

# Weibull AFT
aft_w <- survreg(Surv(time, status) ~ age + sex, data = lung,
                 dist = "weibull")
summary(aft_w)

# Log-normal AFT
```

```r
aft_ln <- survreg(Surv(time, status) ~ age + sex, data = lung,
                  dist = "lognormal")

# flexsurv alternative
flex_w <- flexsurvreg(Surv(time, status) ~ age + sex, data = lung, dist = "weibull")
flex_w

# AFT interpretation
exp(coef(aft_w)[-1])   # acceleration factors (time ratio)
```

Output & Results

`survreg()` returns AFT coefficients on the log-time scale plus the scale parameter; `flexsurvreg()` provides equivalent estimates with cleaner prediction interfaces. Exponentiating the slope coefficients gives acceleration factors interpretable as time ratios.

Interpretation

A reporting sentence: "A Weibull AFT model estimated that each additional year of age accelerated the event by a factor of 0.98 (95 % CI 0.97 to 1.00) — equivalently, a 2 % shorter expected time per year — and females had 30 % longer expected survival than males (acceleration factor 1.30, 95 % CI 1.06 to 1.59)." Choose the AFT-vs-PH framing based on the audience: clinicians who think in survival times prefer acceleration factors, while those used to hazard ratios prefer PH.

Practical Tips

- Weibull AFT is equivalent to Weibull PH up to a fixed transformation; choose the parameterisation that matches the audience.
- Use AIC across distributions (exponential, Weibull, log-Normal, log-logistic, generalised gamma) to select the family; AICs differ substantially when the hazard shape varies.
- Check goodness of fit with Cox-Snell residuals (`flexsurv::resid()`) — a Cox-Snell residual against exp(1) should follow the unit exponential.
- The log–log survival plot is the standard diagnostic for Weibull suitability; clear non-linearity rules it out and motivates a different family.
- For non-monotone hazards (log-Normal, log-logistic), AFT is mandatory — there is no PH analogue without a time-varying coefficient.
- For very small samples, AFT models are more efficient than Cox because the parametric structure adds information; for large samples with PH, Cox is competitive without distributional assumptions.

R Packages Used

`survival` for `survreg()` with multiple AFT families; `flexsurv` for `flexsurvreg()` with cleaner prediction and richer family support including generalised gamma; `eha` for additional parametric extensions; `rstpm2` for spline-based parametric models that bridge AFT and Royston-Parmar.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation